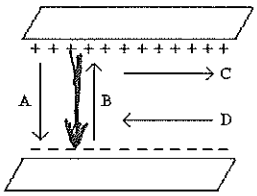
Practice Problems
Calculus Physics II

Chapter 40 Capacitors and Dielectrics

1) Which of the arrows shown in the figure below represents the correct direction of the electric field between the two metal plates?



$\vec{E}$-field is from positive to negative.

A) A
B) B
C) C
D) D
E) none of the above

2) Which of the following will increase the capacitance between the plates of a parallel plate capacitor?

$\kappa = \frac{C}{C_0}$

$C = \kappa C_0$

A) Increase the charge on the plates.
B) Increase the potential between the plates.
C) Introduce a dielectric material between the plates.
D) Decrease the potential between the plates.
E) none of the above

3) When a dielectric material is introduced between the plates of a parallel plate capacitor the capacitance increases by a factor of 4. What is the dielectric constant of the material introduced between the plates?

$C = \kappa C_0$

A) 1/4
B) 0.4
C) 4
D) 2
E) None of the other choices is correct.

4) When the magnitude of the charge on each plate of an air-filled capacitor is 4 μC, the potential difference between the plates is 80 V. What is the capacitance of this capacitor?

$q = CV$ $\quad C = \frac{q}{V} = 5 \times 10^{-8}\,F$

5) What charge appears on the plates of a 2.0-μF capacitor when it is charged to 100 V?

$q = CV = 2 \times 10^{-4}\,C$

6) A 5.0-μC positive charge is moved from a negative to a positive plate of a parallel plate capacitor. In moving this charge 0.30 mJ of energy is used. What is the potential difference between the plates of this capacitor?

$W = -\Delta PE$

$\Delta PE = q\Delta V$

$\Delta V = \frac{\Delta PE}{q} = 60\,V$

7) The potential difference between the plates of a parallel plate capacitor with the plate separation of 6 cm is 60 V. What is the electric field between the plates of this capacitor?

$\Delta x = 6 \times 10^{-2}\ m$ | $\Delta V = |\vec{E}| \Delta x$

$\Delta V = 60\ V$ | $|\vec{E}| = \frac{\Delta V}{\Delta x} = 1{,}000\ V/m$

8) If there is a parallel plate capacitor with plate separation of 4.0 cm has a plate area of 0.04 m^2 and a charge on each plate of 2 nC, what is the Electric Field between the plates of this capacitor with air between these plates?
$\varepsilon_o = 8.85 \times 10^{-12}\ C^2/Nm^2$

$|\vec{E}| = \frac{\sigma}{\varepsilon_o} = \frac{q}{\varepsilon_o A} = 5{,}650\ V/m$

9) A capacitor has a voltage of 391 V applied across its plates, and then the voltage source is removed. What is the potential difference across its plates if the space between them is then filled with mica, having a dielectric constant of 5.4?

$\kappa = \frac{V_o}{V}$ | $V = \frac{V_o}{\kappa} = \frac{391\ V}{5.4} = 72\ V$

10) A parallel plate capacitor with plate separation of 4.0 cm has a plate area of 0.06 m^2. What is the capacitance of this capacitor if a dielectric material with a dielectric constant of 2.4 is placed between the plates?
$\varepsilon_o = 8.85 \times 10^{-12}\ C^2/Nm^2$

$\Delta x = 0.04\ m = d$
$A = 0.06\ m^2$
$\kappa = 2.4$

$C = \kappa\left(\frac{\varepsilon_o A}{\Delta x}\right) = 3.2 \times 10^{-11}\ F$

11) A 20-μF capacitor has a charge of 60 μC. How much energy is stored in this capacitor?

$C = 20 \times 10^{-9}\ F$
$q = 60 \times 10^{-6}\ C$

need ΔV

$\Delta V = \frac{q}{C} = 3\ V$

$Energy = \frac{1}{2} C V^2 = 9 \times 10^{-5}\ J$

12) A camera flash requires 2 mJ of electrical energy to produce the correct amount of light, if the capacitor is charged to 6 V, what must be the capacitance of the capacitor?

$Energy = \frac{1}{2} C V^2$

$C = \frac{2(Energy)}{V^2} = 1.1 \times 10^{-4}\ F$

Answers

1) A
2) C
3) C
4) $C = 5 \times 10^{-8}$ F
5) $q = 2 \times 10^{-4}$ C
6) $\Delta V = 60$ V
7) 1,000 V/m
8) 5650 V/m
9) $\Delta V = 72$ V
10) 3.2×10^{-11} F
11) 9×10^{-5} J
12) 1.1×10^{-4} F